

# Boris Nikolic

Software Developer · backend-focused full-stack  
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## SUMMARY

Software developer building platforms and shipping software with AI agents in the loop. Two years of professional experience in software development, backend-focused full-stack: Odoo modules and data systems on PostgreSQL, Go and Python services, React, Angular, and Next.js frontends, MVC frameworks (Laravel, Django, Spring Boot) on client and university work, and systems and networking code in C and C++. Finishing a BSc in Software Engineering (2023 – 2027). Runs a self-hosted home lab: Ubuntu servers, iptables firewalls, automated SSH hardening, Grafana and Prometheus.

## CORE COMPETENCIES

- **Backend services and APIs:** Go, Python, NestJS; REST design, auth, gateway-fronted services
- **Odoo modules and PostgreSQL:** ERP modules, query optimization, triggers, reporting over large datasets
- **ERP integration and data replication:** Odoo data into analytical warehouses, built for reliable delivery
- **AI-assisted engineering:** Claude Code, role prompts, plugins, MCP servers for pipelines and infrastructure
- **Full-stack delivery:** React, Angular, and Next.js frontends; Express, Laravel, Django, and Spring Boot backends; JWT auth, payments, admin tooling
- **Linux and self-hosting:** Ubuntu servers, iptables firewalls, SSH hardening, monitoring, Docker Compose
- **Working with stakeholders:** Requirements gathered directly, trade-offs explained in plain language, modules and features shipped, tested, and tailored to the production VPS

## EXPERIENCE

**Software Developer, Fairphone (contract)** 2025 – Present

- Odoo modules and data systems for supply-chain and sustainability operations.
- Built Odoo modules (Python, PostgreSQL) for sustainability and procurement operations, replacing manual, hand-kept tracking with automated workflows and reporting over large datasets.
  - Designed and built the replication of ERP data into analytical warehouses, with reliable delivery as the core requirement.
  - Designed the architecture of a customer-facing web shop on top of Odoo: an API gateway in front of the ERP and separate frontend services.
  - Worked directly with stakeholders: gathered requirements from the people doing the work, explained technical trade-offs in plain language, and shipped modules and features end to end, tested and tailored to the production VPS.
  - Work with AI agents in the delivery loop: Claude Code and Codex with per-repository instructions, role prompts, plugins, and MCP servers for pipeline and infrastructure automation.

Client-owned, licensed software; details limited by agreement.  
Over 50 personal and client projects since school; most are private or under agreements not to discuss them. Client work is described at the level the engagements allow.

## PROJECTS

### Toy Store, full-stack e-commerce

Go, Gin, Next.js 14, PostgreSQL 16, Redis 7, Nginx, Stripe, Docker Compose  
E-commerce platform with dual-token JWT authentication (refresh tokens revocable via Redis), cart and wishlist, two-step Stripe PaymentIntent checkout, order management with line-item snapshots, admin dashboard, and a cache-aside Redis catalog with graceful degradation. [github.com/bok1c4/toy\\_store](https://github.com/bok1c4/toy_store)

### RPG campaign manager (university coursework)

Java 21, Spring Boot 3.4, Spring Security (JWT), PostgreSQL 16, React 18, Docker Compose  
Final project for the Internet Software Architecture course: REST API with JWT access and refresh tokens and role-based authorization, CRUD for campaigns, characters, sessions, and items, React SPA, Docker Compose. [github.com/bok1c4/rpg-campaign-manager](https://github.com/bok1c4/rpg-campaign-manager)

### Control Management System Platform

Go, Next.js, React, TypeScript, PostgreSQL  
Multi-role CMS web application with authentication, order management, revenue reporting, a notification system, and role-based access control.

### Systems and networking in C/C++

C, C++20, POSIX sockets, PostgreSQL, GPG, AES-256-GCM, CMake  
A multi-hop HTTP forward proxy on native sockets with layered encryption in progress, a multi-client HTTP socket server, and a terminal password manager storing secrets in PostgreSQL under a hybrid GPG + AES-256-GCM scheme.  
[github.com/bok1c4/HTTP-Proxy](https://github.com/bok1c4/HTTP-Proxy)

## Home lab

Ubuntu, Forgejo, iptables, SSH, Grafana, Prometheus

Self-hosted Ubuntu servers with a Forgejo git forge, home network and custom iptables firewall, SSH hardening automated end to end with an own script, Grafana and Prometheus monitoring, and local blockchains for experiments.

## SKILLS

- **Languages:** Go, Python, TypeScript, C/C++, Rust, SQL; also JavaScript, PHP, Java
- **Backend and APIs:** Odoo module development, Go (Gin, chi), NestJS / Node.js, Express, REST API design, JWT auth and RBAC; also Laravel, Django, Spring Boot, Stripe integration, API gateway pattern
- **Data and storage:** PostgreSQL, ERP data replication, Redis, Query optimization; also MongoDB, SQLite, Analytical warehouses, Prisma, TypeORM, Mongoose, Spring Data JPA
- **Messaging and workflows:** RabbitMQ, Event-driven design, Scheduled jobs, Reliable delivery
- **Frontend:** React, Next.js (SSR), Angular, Tailwind CSS; also React Router, Redux Toolkit, Vite, Zustand, Zod, Axios
- **Infrastructure and operations:** Linux, Ubuntu servers, Bash scripting, Docker and Docker Compose, Nginx, Grafana, Prometheus; also GitHub Actions, GitLab CI, Forgejo (self-hosted), Vercel, Makefiles
- **Security and networking:** iptables firewalls, SSH hardening (automated), Socket programming (C/C++), Nmap, Kali Linux; also HTTP servers and proxies from scratch, PGP handling, Data encryption (AES-256-GCM, GPG)
- **Architecture and practices:** Distributed systems, Domain modeling, Layered (clean) architecture, MVC (Laravel, Spring MVC, Django), Caching strategies, Fault-tolerant data pipelines; also Automated tests (Jest)
- **AI-assisted workflow:** Claude Code, Codex, CLAUDE.md and role prompts, superpowers plugin, engineering plugin, MCP servers; also qodo-standards plugin, Local LLMs
- **Tools:** Neovim (LazyVim), tmux, lazygit, Git; also Lua (Neovim config)

## EDUCATION

**Bachelor of Software Engineering** · Enrolled 2023, finishing by the end of 2027

Path: robots, middle school → linux in a vm → hack the box, high school → software engineering, university → frontend developer → software developer.